Problem Chosen	2026 MCM/ICM Summary Sheet	Team Control Number
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Summary

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Keywords: keyword1; keyword2

Contents

1 Introduction

1.1 Problem Background

Illegal wildlife trade (IWT) is a transnational environmental crime that accelerates biodiversity loss, undermines governance, and threatens ecosystem services. Global syntheses of seizure records show that wildlife trafficking affects thousands of taxa and spans many jurisdictions. They also reveal that trafficking networks adapt rapidly to enforcement pressure through route switching, concealment, and shipment fragmentation. In parallel, operational intelligence from multinational enforcement actions indicates that contemporary IWT networks increasingly exploit high-throughput logistics systems, including international air cargo, postal and express parcels, and digitally mediated trading channels. From both policy and operational perspectives, IWT presents two persistent challenges. First, available data are inherently biased because seizures reflect not only the underlying illegal flows but also variation in detection effort. Thus, a region with fewer seizures does not necessarily experience less trafficking. Second, the system is dynamic and strategic: targeted interdiction at a single border point can displace flows to alternative routes or modes rather than eliminate them. Effective interventions therefore need to be designed at the network level and evaluated under plausible displacement scenarios. These features motivate a data-driven, risk-based approach that can prioritize scarce enforcement resources, quantify expected impact, and remain robust to adaptation.

1.2 Restatement of the Problem

This project asks for a data-driven, five-year intervention plan that can produce a clear and substantial reduction in illegal wildlife trade (IWT), and for a recommendation of a credible client able to implement the plan at scale. The proposed solution should:

- select a realistic client with the international organizations, multinational cooperations, and institutional incentives to deliver the project;
- explain the fit between the client and the project, drawing on relevant literature, operational considerations, and quantitative evidence;
- set out any additional powers, resources, data access, and inter-agency or cross-border coordination needed for effective implementation;
- estimate the project's impact on IWT using transparent modeling assumptions, a clear analytical method, and measurable performance indicators;
- assess implementation feasibility and likelihood of success, and carry out a context-specific sensitivity analysis to highlight key factors that could significantly strengthen or weaken results.

The problem also allows IWT to be modeled within a broader system. If this framing is used, the benefits and limitations of the model, as well as its potential unintended consequences, must be clearly stated and justified.

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1.4 Our Work

2 Model Preparation

2.1 Assumptions

Assumptions 1: For all candidate clients, all indicators are measured within a consistent time window. Each indicator is strictly positive and has been oriented such that larger values indicate better performance and correspond to higher scores.

Assumptions 2:

2.2 Notations

Table 1: Notations used in the entropy weight model

Symbol	**Description**
m	The number of candidate institutions.
n	The number of evaluation indicators; in this study $n = 13$.
$\mathbf{X}$	The original data matrix of size $m \times n$.
x_{ij}	The raw value of institution i under indicator j.
i	The index of institution, $i = 1, 2, \ldots, m$.
j	The index of indicator, $j = 1, 2, \ldots, n$.
p_{ij}	Column-normalized value.
e_j	The information entropy.
d_j	The degree of diversification: $d_j = 1 - e_j$.
w_j	Entropy weight.
S_i	Comprehensive score.
k	Index of category.
M_k	The set of institutions belonging to category k.
$\lvert M_k\rvert$	The number of institutions in M_k.
$\overline{S}_k$	The average score of category k.
$\ln(\cdot)$	Natural logarithm.

2.3 Data Collection and Visualization

3 Model Establishment

3.1 Model I: Client Selection using Entropy Weighting and Decision Tree

We establish a two-stage framework to select a ideal and realistic client for the proposed five-year intervention. In Stage 1, we apply the Entropy Weight Method to identify the most suitable client category from three options: *International Organizations*, *Government Agencies*, and *Technology Companies*. Then in Stage 2, given that the candidates are more complex and numerous, we use an decision tree to select a specific institution within the chosen category.

3.1.1 Indicator System

A total of $n = 13$ indicators are designed and organized into four dimensions: *Authority*, *Resources*, *Coordination*, and *Strategy*. Each indicator is positively oriented, meaning that larger values indicate stronger capability or better fit. Specifically:

Table 2: Indicator system for Model I

Dimension	**Code**	**Description**
Authority (A)	A1	Annual enforcement-related budget.
Authority (A)	A2	The number of enforcement and supervision personnel.
Authority (A)	A3	The number of policies or standards released.
Authority (A)	A4	Population coverage of legal influence.
Resources (B)	B1	Annual total budget.
Resources (B)	B2	The number of technical personnel.
Resources (B)	B3	The number/scale of relevant databases.
Resources (B)	B4	The number of patents.
Resources (B)	B5	The number of publicly available models/reports.
Coordination (C)	C1	The number of international cooperation agreements.
Coordination (C)	C2	The number of cross-border joint operations/actions.
Strategy (D)	D1	Policy keyword matching degree.

3.1.2 Stage 1: Category Selection via Entropy Weight Method

Let m denote the number of candidate institutions, and let $\mathbf{X} = (x_{ij})_{m\times n}$ be the raw indicator matrix, where x_{ij} represents the observed value of institution i on indicator j ($j = 1, \ldots, 13$). In this matrix, each column corresponds to an evaluation indicator and each row to a specific institution. By construction of the indicator system and the accompanying data protocol, all entries of $\mathbf{X}$ are strictly positive and have already been oriented so that larger values uniformly indicate stronger capability or better suitability.

Step1: Normalization.

Because the 13 indicators differ in magnitude and units, directly aggregating x_{ij} would be inappropriate. To make institutions comparable within each indicator, we apply a column-wise proportion normalization:

$$p_{ij} = \frac{x_{ij}}{\sum_{i=1}^{m} x_{ij}}, \qquad i = 1, \ldots, m, \;\; j = 1, \ldots, n. \tag{1}$$

This transformation converts each original value x_{ij} into a proportion p_{ij} for indicator j. By construction, $\sum_{i=1}^{m} p_{ij} = 1$ for every j, so each indicator column becomes a probability-like distribution over institutions. Intuitively, a larger p_{ij} indicates that institution i contributes more to the overall level of indicator j among all candidates, reflecting a relatively stronger performance on that indicator.

Step2: Entropy of each indicator.

The entropy weight method (EWM) assigns indicator weights in an objective way, based on how much information is contained in their observed distributions. For indicator j, its entropy is defined as

$$e_j = -\frac{1}{\ln m} \sum_{i=1}^{m} p_{ij} \ln(p_{ij}), \tag{2}$$

where the factor $1/\ln m$ ensures that $e_j \in [0, 1]$. Here, entropy describes how even or uneven the distribution $p_{1j}, \ldots, p_{mj}$ is. If the p_{ij} values are almost the same across institutions, the indicator carries little information for comparison; the distribution is close to uniform and e_j is near 1. In contrast, if the indicator differs a lot across institutions and only a few candidates show clear advantages, the distribution is more uneven and e_j becomes smaller. Thus, indicators with lower entropy are seen as more useful for telling institutions apart and are given larger weights in EWM.

Step3: Diversification degree and entropy weight.

To transform entropy into a direct measure of “useful variation,” we define the diversification degree as

$$d_j = 1 - e_j. \tag{3}$$

A larger d_j indicates that indicator j displays greater dispersion and thus a stronger ability to distinguish among institutions. The entropy weight is then obtained by normalizing the diversification degrees:

$$w_j = \frac{d_j}{\sum_{j=1}^{n} d_j}. \tag{4}$$

This construction guarantees $\sum_{j=1}^{n} w_j = 1$ and ensures that the weights are determined solely by data-driven variability rather than subjective preference. Consequently, more discriminative indicators receive higher weights, whereas indicators that are nearly uniform across the candidate set receive lower weights.

Step 4: Institutional composite score.
After the weights are determined, each institution's overall performance is calculated as a weighted sum of its normalized indicator proportions:

$$S_i = \sum_{j=1}^{n} w_j, p_{ij}. \tag{5}$$

The resulting S_i serves as the institution's aggregated competitiveness score across all 13 indicators, with each indicator's contribution scaled by its information content. Since all indicators are positively oriented, a larger S_i corresponds to stronger overall suitability within the evaluation framework.

Step5: Category-level score and selection.
Since the primary goal of Stage 1 is to select the most appropriate *client category* (rather than a specific institution), institutions are grouped into predefined categories. Let M_k denote the set of institutions belonging to category k, and let $|M_k|$ be the number of institutions in that category. The category-level score is defined as the within-category mean:

$$\overline{S}_k = \frac{1}{|M_k|} \sum_{i \in M_k} S_i. \tag{6}$$

This aggregation produces a representative score for each category, reflecting the average institutional strength under the same weighting scheme. The category with the largest $\overline{S}_k$ is selected as the target client type for the next stage, which allows the subsequent decision-tree screening to focus on the most promising organizational context for implementation.

Step 5: Category-level score and selection. Because Stage 1 aims to identify the most suitable *client category* rather than a single institution, institutions are first grouped into predefined categories. Let M_k be the set of institutions in category k, and let $|M_k|$ denote the number of institutions in that category. The category-level score is defined as the average of the institutional scores within the category:

$$\overline{S}k = \frac{1}{|M_k|} \sum i \in M_k S_i. \tag{7}$$

This produces a representative score for each category, capturing its average institutional strength under a common weighting scheme. The category with the highest $\overline{S}_k$ is chosen as the target client type for the next stage, so that the subsequent decision-tree screening focuses on the most promising organizational context for implementation.

3.1.3 Stage 2:

3.2 Model II

3.3 Model III

4 Result and Discussion

5 Model Evaluation

5.1 Sensitivity Analysis

6 Conclusion

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